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A comprehensive model for Doppler spectra in thermal atomic vapour

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Abstract

We theoretically and experimentally investigate the Doppler-broadened absorption spectrum of the D_2-line ($5^2S_{1/2} \rightarrow 5^2P_{3/2}$) in thermal (room-temperature) Rb vapour over a wide range of probe intensity, probe diameter, and vapour cell temperature to understand the effect of hyperfine pumping and transit-relaxation on the absorption lineshape. We present a relatively simple but comprehensive five-level rate equation model which incorporates optical pumping of the atomic population into the other 'dark' ground hyperfine level (leading to absorption saturation), including contributions of the three closely spaced (within one Doppler linewidth) hyperfine levels in the excited state $5^2P_{3/2}$ (allowed by the dipole transition selection rules). D_2-line transmission spectra predicted by our model show excellent agreement (rms error $<5\%$) with the experimental data for a wide range of probe intensities (from $\sim 0.001 I_{sat0}$ all the way up to $\sim 10 I_{sat0}$, where I_{sat0} is the usual two-level saturation intensity for the respective atomic transition), vapour cell temperatures (24.2 °C–53.5 °C) and beam diameters ($1/e^2$ width $\sim 1-6$ mm). Our model also takes into account the finite transit time of atoms across the probe beam cross-section, and correctly predicts the dependence of effective saturation intensity on probe diameter (up to $\sim 10\times$ lower saturation intensity as compared to I_{sat0} for few-mm beam sizes). This *ab initio* five-level model can thus predict accurate Doppler lineshapes for any given experimental parameter set for a linearly polarized probe without any fitting parameters, and can be easily applied to any other atomic system by an appropriate change of the atomic constants in the rate equations.

Keywords: hyperfine-pumping, transit-relaxation, absorption saturation

(Some figures may appear in colour only in the online journal)

1. Introduction

Doppler-broadened absorption is the simplest spectroscopy technique to study the internal structure of atoms without perturbing the level populations [1, 2]. Although limited in helping discern the entire complicated hyperfine structure, Doppler-broadened spectra still reveal some important information such as the population distribution in different atomic states, vapour pressure, saturation intensity, etc [3–7]. Rb vapour is a much preferred system for various quantum and nonlinear optical interactions at low powers [3, 8, 9], due to its high vapour pressure and resulting atomic density at room temperature ($\sim 10^{16}$ atoms m^{-3}) along with the large oscillator strength of the transitions, with applications in precision spectroscopy and metrology [8, 10]. In addition, due to the highly dispersive nature of narrow-linewidth transitions, it provides a much preferred slow-light medium with applications like light storage, light-drag enhancement, etc [11, 12]. The spectral range of Rb D-line transitions lies in the visible/near-IR region at which high-power tunable lasers with narrow linewidth (<1 MHz) are commercially available. Specifically, the Rb D_2-line at $\sim$780.25 nm, comprising both open and closed transitions, has been employed in various applications in laser-frequency locking, laser-cooling, metrology, interferometry, magnetometry, etc [2, 13–15]. Thus it is extremely important to

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study the Doppler absorption spectrum in detail and quantify its dependence on various experimental parameters used in practice.

For a sufficiently weak probe ($\lesssim 0.01 I_{\rm sat0}$), some extensive and useful theoretical and experimental studies of the absorptive and dispersive properties of Doppler-broadened D lines of thermal alkali vapours have been reported in [3, 16, 17], based on a simple two-level susceptibility model. Here, we present a comprehensive theoretical model to accurately predict the Doppler absorption spectrum of the D_2-line of Rb near room temperature by taking hyperfine pumping into account, and experimentally verify our model over a wide range of parameters through detailed measurements. The study of hyperfine pumping in multi-level systems like Rb can provide a better insight into precision Doppler-free spectroscopic techniques as well, such as saturated absorption spectroscopy, polarization spectroscopy, electromagnetically induced transparency, etc [15, 18, 19]. Figure 1(a) shows the hyperfine energy levels with their frequency separation for the Rb D_2-line for both the stable isotopes, ^{85}Rb and ^{87}Rb [3]. The Doppler-broadened spectrum contains four transmission dips (two for each isotope, see figure 3), each dip comprising one closed and two open hyperfine transitions (see table 1). Closed hyperfine transitions are those for which only one decay channel from the excited state (back to the initial ground state) is allowed by the electric-dipole selection rule ($\Delta F = 0, \pm 1$) [1]. Such atomic transitions can be modelled as ideal two-level systems. However, in open hyperfine transitions, atoms optically pumped from one hyperfine ground state to the excited state are allowed to decay spontaneously to another different hyperfine ground state. Thus, in multi-level systems such as Rb, a simple two-level susceptibility model is found to be inadequate to calculate the absorption spectrum accurately [3, 18], even for probe intensities as low as $\gtrsim 0.01 I_{\rm sat0}$ (see figures 3 and 4). To overcome this limitation of the two-level model, open transitions can be modelled as three-level systems that include one ground and one excited state coupled via the probe laser along with another ground state into which atoms get transferred via hyperfine pumping [4, 20]. For example, a three-level rate equation model has been successfully used to calculate the Doppler-broadened absorption spectrum of the Cs D_1-line [4]. Here, we further extend this to a five-level rate equation model for the Rb D_2-line, to include contributions of the three closely spaced (within one Doppler linewidth) hyperfine levels (allowed by the dipole selection rules $\Delta F = 0, \pm 1$) in the excited state $5P_{3/2}$ (see figure 1). Also, to the best of our knowledge, there is no detailed comparison yet reported between the *ab initio* (i.e. without any fitting parameters) theoretically calculated and experimentally observed Doppler transmission spectrum of the Rb D_2-line with varying probe intensity, beam diameter, and vapour cell temperature. Our five-level model takes into account the effect of hyperfine pumping occurring within the transit time of the atoms across the beam cross-section, and shows excellent agreement with the experimentally observed spectra for a wide range of probe intensity, varying from $\sim 0.001 I_{\rm sat0}$ to $\sim 10 I_{\rm sat0}$ (see figures 3 and 4), where $I_{\rm sat0}$ is the two-level saturation intensity, an atomic constant that depends only on the spontaneous decay rate [13]. Hyperfine pumping also leads to a dependence of the effective saturation intensity on the probe beam diameter (due to the interplay of the optical pumping and atomic transit timescales), as has been measured in [4, 21, 22]. However, an accurate quantitative model for the same has not been reported yet. Our five-level model provides the correct dependence of saturation intensity on probe beam diameter, and we experimentally verify this for various different vapour cell temperatures. We observe up to $10\times$ reduction in the saturation intensity as compared to $I_{\rm sat0}$ (~ 1.6 mW cm^{-2} [23, 24] for the Rb D_2-line) for beam diameters up to ~ 7 mm (see table 3).

2. Theory and calculations

There are two different approaches mainly used to calculate absorption spectra: classical and semiclassical, both giving similar results in terms of accuracy [4, 18, 20, 25, 26]. In the classical approach, absorption is computed as a quantity proportional to the steady-state population difference between the relevant atomic energy levels by solving coupled rate equations [4, 18, 27]. In the semiclassical or density-matrix approach, the imaginary part of calculated susceptibility provides the absorption [20, 25]. However, it becomes quite tedious and computationally resource intensive to incorporate all the degenerate magnetic sublevels (in the absence of magnetic field) of a multi-level system like Rb in the density-matrix model (although, to mitigate this somewhat, a reduced density-matrix formalism is introduced in [25, 28]). Therefore, we employ a simpler five-level rate equation model for the Rb D_2-line, which is demonstrated via experimental confirmation to work well for probe intensities all the way up to $\sim 10 I_{\rm sat0}$. In the presence of a magnetic field, however, a full multi-level rate equation model or density-matrix approach will be required to include all the non-degenerate magnetic sublevels [4, 18, 26, 29, 30].

2.1. Five-level rate equation model for Rb D_2-line

Figure 1(b) shows the schematic of the five-level model, which comprises two ground states $|a\rangle$, $|c\rangle$, and three excited states $|b_1\rangle$, $|b_2\rangle$, and $|b_3\rangle$. Here, $|a\rangle$ is the 'bright' ground state coupled to the three excited states via the probe laser and $|c\rangle$ is the 'dark' ground state into which population gets transferred via hyperfine pumping. For example, in the Rb D_2-line, the three allowed optical transitions from the lower hyperfine ground state of ^{87}Rb are $F_{\rm g} = 1 \rightarrow F_{\rm e} = 0, 1, 2$, according to electric-dipole selection rules $\Delta F = 0, \pm 1$. Here, $F_{\rm g} = 1$ is labelled as $|a\rangle$ and $F_{\rm e} = 0, 1, 2$ are labelled as $|b_1\rangle$, $|b_2\rangle$, and $|b_3\rangle$ respectively (see figure 1(a)). The other hyperfine ground state of ^{87}Rb, $F_{g'} = 2$, referred to as $|c\rangle$, exchanges population with the excited states via spontaneous decay and with $|a\rangle$ via transit relaxation (discussed in section 2.2).

The following rate equations (1) to (3) govern the change of atomic population with time among these five levels [4, 18, 31]:

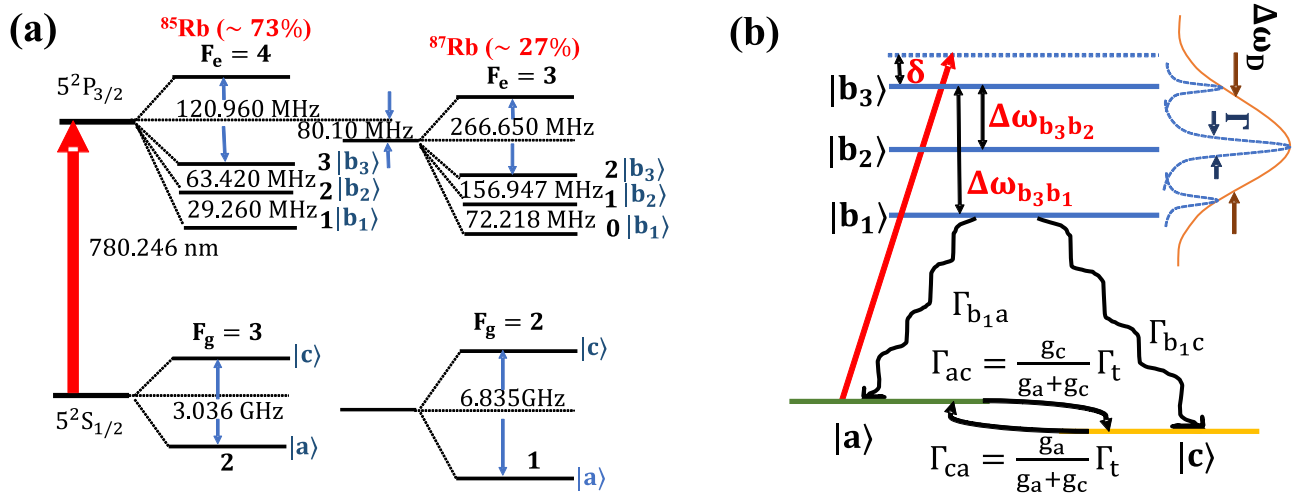


Figure 1. (a) Hyperfine-level scheme and frequency separation for the Rb D_2-line transitions for both isotopes (^{85}Rb and ^{87}Rb) and (b) schematic for the five-level model indicating absorption in the presence of hyperfine pumping.

Table 1. The relative coupling strength of the three dipole-allowed hyperfine transitions from the upper hyperfine ground state of (a) ^{87}Rb, and (b) ^{85}Rb; and lower hyperfine ground state of (c) ^{85}Rb, and (d) ^{87}Rb (corresponding to dips (i)–(iv) of the transmission spectrum respectively, see figure 3) [14, 23, 24]:

(a)		(b)	
$S_{2,0}$	0	$S_{3,1}$	0
$S_{2,1}$	1/20	$S_{3,2}$	5/63
$S_{2,2}$	1/4	$S_{3,3}$	5/18
$S_{2,3}$	7/10	$S_{3,4}$	9/14
(c)		**(d)**	
$S_{2,1}$	3/10	$S_{1,0}$	1/6
$S_{2,2}$	7/18	$S_{1,1}$	5/12
$S_{2,3}$	14/45	$S_{1,2}$	5/12
$S_{2,4}$	0	$S_{1,3}$	0

$$\dot{n}_a = \sum_{i=1,2,3}\left(\frac{B_{b_i a} I(n_{b_i} - n_a g_{b_i}/g_a)}{1+4\left(\frac{\delta_{\rm pr}+\Delta\omega_{b_3,b_i}}{\Gamma}\right)^2} + \Gamma_{b_i a} n_{b_i}\right) - \Gamma_{ac} n_a + \Gamma_{ca} n_c \quad (1)$$

$$\dot{n}_{b_i} = -\frac{B_{b_i a} I(n_{b_i} - n_a g_{b_i}/g_a)}{1+4\left(\frac{\delta_{\rm pr}+\Delta\omega_{b_3,b_i}}{\Gamma}\right)^2} - (\Gamma_{b_i a} + \Gamma_{b_i c}) n_{b_i} \quad (i = 1,2,3) \quad (2)$$

$$\dot{n}_c = \sum_{i=1,2,3} \Gamma_{b_i c} n_{b_i} + \Gamma_{ac} n_a - \Gamma_{ca} n_c. \quad (3)$$

These rate equations are solved under a constraint equation (4), which ensures the conservation of total atomic density n (which is a function of cell temperature T [3, 32]):

$$n(T) = n_a + n_{b_1} + n_{b_2} + n_{b_3} + n_c. \quad (4)$$

This model includes the effect of Doppler-broadening in the form of effective probe detuning, $\delta_{\rm pr} = \delta - kv$, where $\delta = \omega - \omega_0$ is detuning of probe frequency (ω) from atomic resonance frequency (ω_0), $k = \frac{2\pi}{\lambda}$ is propagation constant of probe and v is the velocity component of the atom along $\vec{k}$. $\Delta\omega_{b_3,b_i}$ is the angular frequency separation between ith excited state, $|b_i\rangle$, and the third excited state $|b_3\rangle$. By definition, $\Delta\omega_{b_3,b_3} = 0$ (where, $\omega_{b_3} - \omega_a \equiv \omega_0$) and the values of $\Delta\omega_{b_3,b_2}$ and $\Delta\omega_{b_3,b_1}$ are given in figure 1(a) [3, 31]. For the Rb D_2-line, the centre frequency of the transition is $\omega_0 = 2\pi \times 384.23$ THz i.e. the centre wavelength is $\lambda = 780.246$ nm. $B_{b_i a} = \frac{\lambda^3 \Gamma_{b_i a}}{4\pi h c \Gamma_{b_3 a}}$ represents the Einstein B coefficient ($\Gamma_{b_i a}$ is the spontaneous decay rate defined below) corresponding to the transition from $|a\rangle \rightarrow |b_i\rangle$, I is the probe intensity, and $g_i = 2F_i + 1$ is the degeneracy of ith hyperfine state (the number of degenerate magnetic hyperfine sublevels) [2, 13, 23].

2.2. Atomic population relaxation

In this five-level model, we take two relaxation processes into account. One is spontaneous (or fluorescence) decay of excited atomic population to any of the two ground states, $|a\rangle$ or $|c\rangle$. The other is transit-relaxation which is effectively an exchange of atomic population between the two ground-states $|a\rangle$ and $|c\rangle$ via atoms moving in and out of the probe beam cross-section. Atom–atom collisions are neglected in our model as at room temperature in the absence of any buffer gas (our vapour cell is vacuum sealed), collision (or pressure) broadening for Rb is nearly four orders of magnitude smaller than lifetime broadening (natural linewidth) [3, 4].

2.2.1. Spontaneous decay rate. The total spontaneous decay rate of the population from ith excited state $|b_i\rangle$ is given as [4]:

$$\Gamma_{b_i} = \Gamma_{b_i a} + \Gamma_{b_i c} = \Gamma, \quad (5)$$

where Γ is full-width at half maximum (FWHM) atomic transition linewidth [3]:

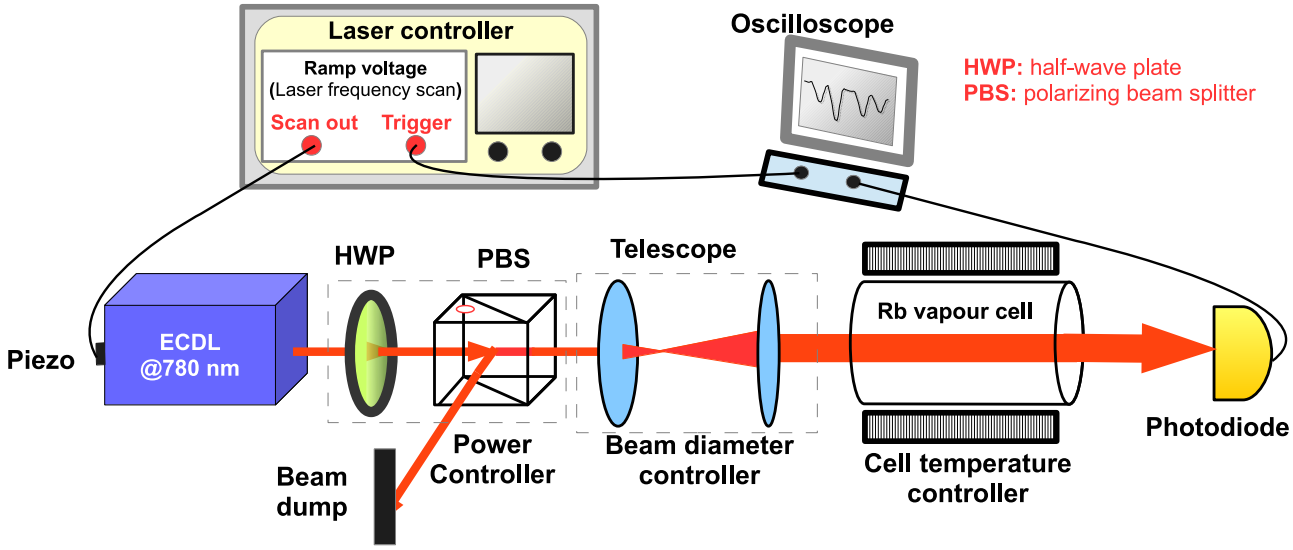


Figure 2. Experimental setup for studying the effect of variation of probe intensity, beam diameter and vapour cell temperature on the Doppler-broadened absorption spectrum of Rb D_2-line.

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}\frac{2J_g+1}{2J_e+1}|\langle J_g||er||J_e\rangle|^2, \tag{6}$$

where $\langle J_g||er||J_e\rangle$ is dipole matrix element corresponding to the transition from state $J_g \rightarrow J_e$. For the Rb D_2-line transition ($J_g = 1/2 \rightarrow J_e = 3/2$), $\Gamma = 2\pi \times 6.06 \times 10^6$ Hz. The decay rates corresponding to the transition from a specific excited hyperfine state F_e (or $|b_i\rangle$) to the hyperfine ground state F_g (or $|a\rangle$) and $F_{g'}$ (or $|c\rangle$) are [14, 33]:

$$\Gamma_{b_i a} = 2\frac{g_a}{g_{bi}}S_{a,b_i}\Gamma \quad \text{and} \quad \Gamma_{b_i c} = 2\frac{g_c}{g_{bi}}S_{c,b_i} \tag{7}$$

which satisfy the relation (5). Here, g_i is the degeneracy of ith hyperfine state, 2 is the degeneracy ratio of excited ($5^2P_{3/2}$, comprising four hyperfine levels) to ground state ($5^2S_{1/2}$, comprising two hyperfine levels). S_{a,b_i} (or S_{c,b_i}) denotes the relative coupling (oscillator) strength of the allowed transitions from the ground state $|a\rangle$ (or $|c\rangle$) to the three excited states (see table 1 for Rb D_2-line).

2.2.2. Transit-relaxation rate. An atom interacting with the probe laser does not achieve its steady-state instantaneously as the optical-pumping time (∼few μs) [26] is comparable to the transit time across the beam. These comparable time scales result in a non-stationary phenomenon which affects the absorption lineshape [18, 26]. This effect can be investigated by varying the optical pumping rate (via probe intensity) and transit time (via probe beam diameter and/or thermal velocity of the atoms i.e. temperature of the vapour cell). Our five-level model takes both these time-scales into account as it includes the pumping term, $B_{b_i a}I$, and the transit-relaxation rates, Γ_{ac} (and Γ_{ca}), governing the effective transfer of atomic population from $|a\rangle \rightarrow |c\rangle$ (and $|c\rangle \rightarrow |a\rangle$). A complete derivation of the transit relaxation rates as given below in equations (8) and (9) has been discussed in [4]. The approxlimation of transit-relaxation modeled as $\Gamma_{ij}n_i$ works well when compared to other models, yielding results close to that from the more complicated exact expression [18, 22, 34]. Atoms are assumed to be entering the probe beam as a statistical mixture of the population in the two hyperfine ground states (equal distribution in all the degenerate magnetic sub-levels), and atoms leaving the probe beam can be present in any of the two hyperfine ground states with the probability governed by following transit relaxation rates [4]:

$$\Gamma_{ac} = \frac{g_c}{g_a+g_c}\Gamma_t \quad \text{and} \quad \Gamma_{ca} = \frac{g_a}{g_a+g_c}\Gamma_t, \tag{8}$$

where, Γ_t is total transit-relaxation rate ($\Gamma_t = \Gamma_{ac} + \Gamma_{ca}$) given as:

$$\Gamma_t = \frac{v_{avg}}{D}, \tag{9}$$

where $v_{avg} = \sqrt{\frac{8k_BT}{\pi m}}$ is the average thermal velocity of the atoms and D equals the FWHM of the probe beam having Gaussian intensity profile (D ∼ 0.58 $1/e^2$ full-width) [35]. In this paper, the beam diameter values that we quote correspond to the $1/e^2$ full-width.

2.3. Calculation of absorption coefficient

For a Doppler-broadened medium like thermal atomic vapour, the net absorption coefficient (α_D) for the probe can be obtained by summing the individual absorption coefficients ($\alpha_{b_i,a}$, $i = 1, 2, 3$) corresponding to the allowed transitions from the ground state $|a\rangle$ to the three excited states $|b_i\rangle$, and integrating over the Maxwell–Boltzmann velocity distribution:

$$\alpha_D(\delta, I) = \int_{-\infty}^{\infty} f(\delta_{pr} - \delta)\sum_{b_i}\alpha_{a,b_i}(\delta, \delta_{pr})d\delta_{pr}, \tag{10}$$

where,

$$f(\delta_{pr} - \delta) = \frac{\lambda}{2\pi^{3/2}v_p}e^{-\left(\frac{\delta_{pr}-\delta}{kv_p}\right)^2} \tag{11}$$

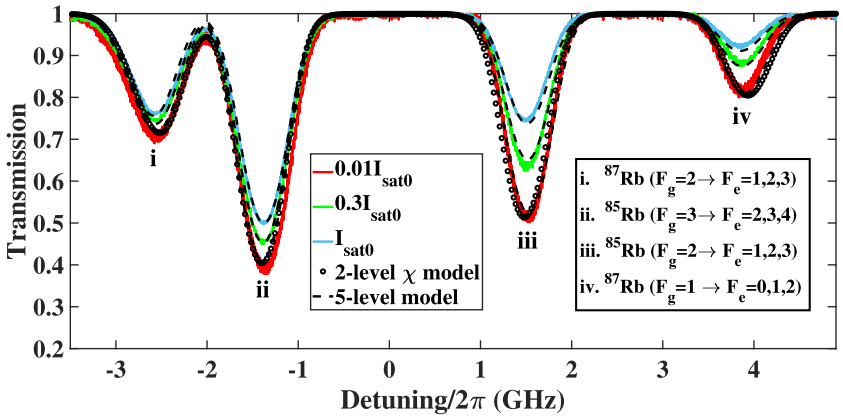


Figure 3. Experimentally observed Doppler-broadened transmission spectra of Rb D_2-line (coloured solid lines) show good agreement with the calculated transmission spectra using our five-level model (black dashed line) at three different probe intensities ($0.01I_{sat0}$, $0.3I_{sat0}$ and I_{sat0}) with a fixed probe beam diameter ∼2.86 mm and vapour cell temperature 24.2 °C. The black dotted line shows the spectrum calculated from the two-level χ-model [3].

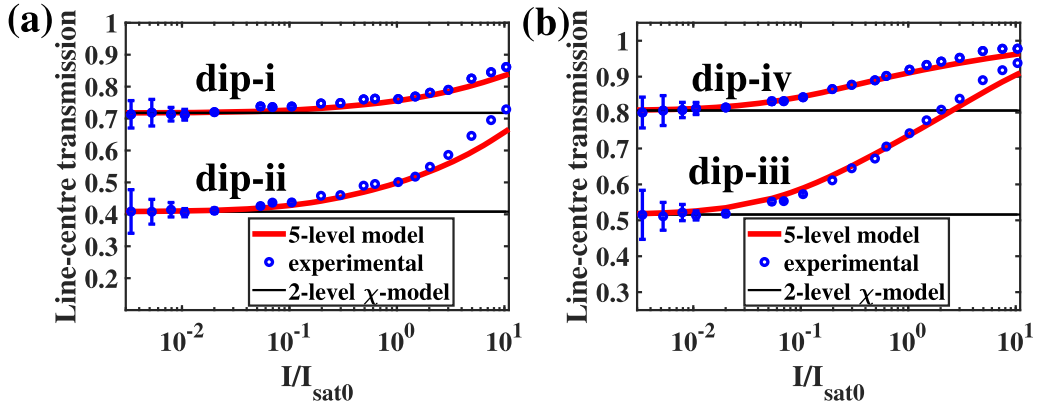


Figure 4. Calculated line-centre transmission vs normalized probe intensity I/I_{sat0}. The five-level model (red solid lines) shows good agreement with experimental data (blue dots) for all four dips of the Rb D_2 line for beam diameter ∼2.86 mm, and cell temperature 24.2 °C. The horizontal black solid line denotes the two-level χ-model [3] which does not take probe intensity into account.

is the Maxwell–Boltzmann velocity distribution function in terms of probe detuning, $v_p = \sqrt{\frac{2k_BT}{m}}$ is the most probable velocity of the atoms [1], and

$$\alpha_{a,b_i}(\delta,\delta_{pr}) = \frac{\lambda^2\Gamma_{b_ia}}{2\pi\Gamma}\frac{1}{1+4\left(\frac{\delta_{pr}+\Delta\omega_{b_3,b_i}}{\Gamma}\right)^2} \times \Delta n(\delta_{pr},I,\delta,t\to\infty) \quad (12)$$

,where, the steady population difference calculated by solving equations (1) to (4) is

$$\Delta n(\delta_{pr},\delta,I,t\to\infty) = \frac{g_{b_i}}{g_a}n_a(\delta_{pr},\delta,I,t\to\infty) - n_{b_i}(\delta_{pr},\delta,I,t\to\infty). \quad (13)$$

The absolute Doppler-broadened transmission spectrum for each dip of the Rb D_2-line can be calculated by using the following expression:

$$T_D(\delta,I) = e^{-\alpha_D(\delta,I)L}, \quad (14)$$

where L is length of the vapour cell.

3. Experimental setup

Figure 2 shows a schematic of the experimental setup used to verify the five-level model predictions. The output of a continuously-tunable external cavity diode laser or ECDL (TOPTICA® DLpro), operating at the Rb D_2-line wavelength (∼780.25 nm), is coupled into a single-mode fiber (Thorlabs® SM780) and collimated back into free-space using a pig-tailed aspheric collimator (Thorlabs® PAF2A-15B) in order to obtain a clean spatial mode i.e. a Gaussian probe beam with a diameter ($1/e^2$ full-width) ∼2.86 mm. A pair of half-wave plate and polarizing beam splitter is used to vary the probe power. A telescope, made with different combinations of two plano-convex lenses, is used to further expand or contract the laser beam to the desired spot size. Afterward, the laser beam is allowed to pass through the Rb vapour cell (Precision Glassblowing®) of dimensions 75 mm (length) × 19 mm (diameter) containing both the isotopes, ^{85}Rb and ^{87}Rb, in their natural abundance of 72.17% and 27.83% respectively [36].

A beam profiler (BeamGage® LT665 from Ophir®) is used to measure the probe beam diameter (second-moment or $1/e^2$

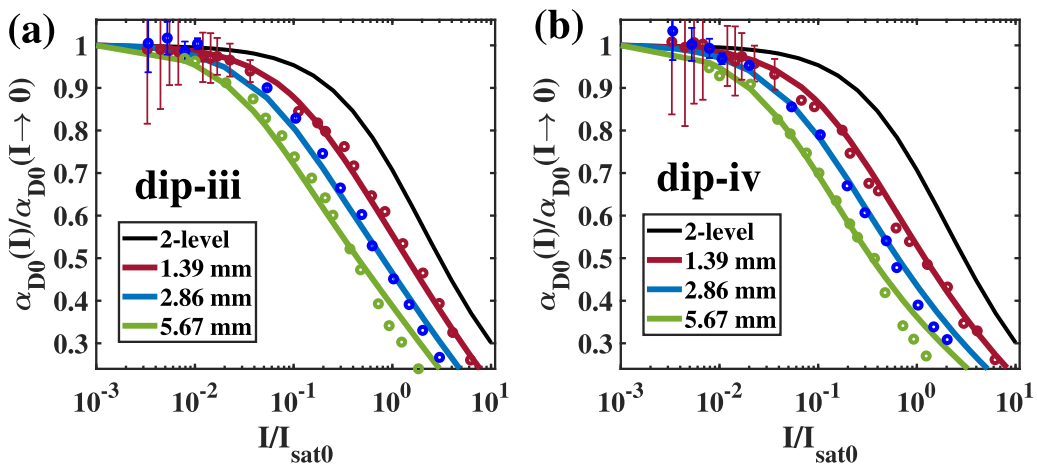


Figure 5. Normalized peak (i.e. line-centre) absorption coefficient α_{D0} vs normalized probe intensity I/I_{sat0} for three different beam diameters $\sim$1.39 mm, 2.86 mm, 5.67 mm (with ellipticity <3%). The five-level model (coloured solid lines) shows fairly good agreement with experimentally observed data (coloured dots) for (a) dip-iii and (b) dip-iv. The black solid line denotes the standard two-level model for a Doppler-broadened medium, i.e., $\frac{\alpha_{D0}(I)}{\alpha_{D0}(I\to 0)} = \frac{1}{\sqrt{1+I/I_{sat0}}}$. The exact values of the vapour cell temperature during these measurements are listed in table 2.

width) after the telescope, just before the entrance of the vapour cell. A temperature controller (Thorlabs® TC 200) is used to set and measure the temperature of the Rb vapour cell (mounted on a heater, Thorlabs® GCH25-75) with an accuracy of $\pm 0.1\,^{\circ}$C. A photodiode, (Thorlabs® PDB210A) is used to detect the probe beam after the Rb vapour cell.

In an ECDL, a diffraction grating (attached with a piezo stage) forms the external cavity of the diode laser, and the laser frequency is scanned by moving it in a controlled manner. We used the analog scan controller of the laser to provide a ramp voltage and a scan offset to the piezo stage to continuously scan the laser frequency over the whole range of Rb D_2-line transitions ($\sim$7 GHz). To avoid any mode-hopping, we scanned the laser frequency at a slow rate of $\sim$0.3 GHz ms^{-1}. An electronic signal provided by the laser scan controller, synced with the piezo ramp signal, is used as an external trigger for the oscilloscope to record the transmission spectrum (figure 2).

4. Results and discussion

4.1. Effect of hyperfine pumping

Figure 3 shows the experimentally observed and theoretically calculated Doppler-broadened transmission spectrum for the Rb D_2-line. It is evident that dips i and ii exhibit reasonably good agreement with the basic two-level χ-model [3] for probe intensity $\lesssim 0.1 I_{sat0}$, whereas dips iii and iv show significant deviation from this model even at much weaker probe intensity ($\gtrsim 0.01 I_{sat0}$), but all the dips have good agreement with the proposed five-level model for probe intensity up to $\sim 10 I_{sat0}$ (see figures 3 and 4). This can be explained by the relative coupling (oscillator) strengths of the one closed transition and two open transitions allowed from the ground-state corresponding to each dip (see table 1). For the first two dips (i) and (ii), the closed hyperfine transition (follows $\Delta F = 1$, i.e. $F_g = 2 \to F_e = 3$ for ^{87}Rb and $F_g = 3 \to F_e = 4$ for ^{85}Rb) has a much larger coupling strength than the two open hyperfine transitions. In this case, the closed transition dominates, thus negating the effect of hyperfine pumping. Whereas for the last two dips (iii) and (iv), the coupling strength of the closed hyperfine transition (follows $\Delta F = -1$, i.e. $F_g = 2 \to F_e = 1$ for ^{85}Rb and $F_g = 1 \to F_e = 0$ for ^{87}Rb) is comparable to the two open hyperfine transitions (see table 1). Hence, hyperfine pumping plays a significant role here.

Figure 4 shows the comparison of calculated and experimentally observed line-centre (minimum) transmission as a function of probe intensity for all four dips (i)–(iv) at cell temperature 24.2 °C and probe beam diameter $\sim$2.86 mm. It is observed that there is a significant increase in the transmission

Table 2. The experimental data for absorption saturation for three different beam diameters shown in figure 5 are fitted to the function $\frac{1}{\sqrt{1+\beta\frac{I}{I_{sat0}}}}$. Here, $\beta = \frac{I_{sat0}}{I_{sat}}$ is a parameter which gives the factor by which the saturation intensity is lowered (due to hyperfine pumping) as compared to the ideal two-level system.

$1/e^2$ diameter	T (°C)	β for dip-iii	β for dip-iv
1.39 mm	27.2	2.35 ± 0.06	2.64 ± 0.08
2.86 mm	24.2	4.04 ± 0.15	4.57 ± 0.2
5.67 mm	29.2	7.30 ± 0.34	8.42 ± 0.39

Table 3. β (factor by which the saturation intensity is lowered due to hyperfine pumping, as defined in table 2) for dip-iv at seven different $1/e^2$ beam diameters. The corresponding values of atomic transit time ($\sim 1/\Gamma_t$) are also listed in order to compare with the typical time-scale of optical pumping ($\sim$10 μs, for $I \sim 0.1 I_{sat0}$) [26, 31]:

$1/e^2$ diameter	β for dip-iv	$1/\Gamma_t$ (μs)
1 mm	2.20 ± 0.04	1.86
2 mm	3.93 ± 0.12	3.73
3 mm	5.54 ± 0.21	5.58
4 mm	7.05 ± 0.30	7.46
5 mm	8.47 ± 0.39	9.34
6 mm	9.78 ± 0.47	11.1
7 mm	10.9 ± 0.54	13.0

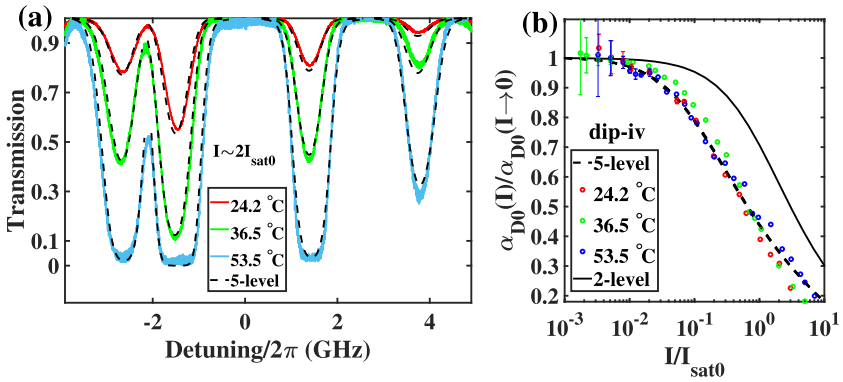


Figure 6. (a) Experimentally observed Doppler-broadened transmission spectra of Rb D_2-line (coloured solid lines) with probe intensity $\sim 2I_{\rm sat0}$, beam diameter $\sim$2.86 mm at three different vapour cell temperatures: 24.2 °C, 36.5 °C and 53.5 °C show good agreement with the five-level model (black dashed lines). (b) Normalized line-centre absorption coefficient $\alpha_{\rm D0}$ vs normalized probe intensity $I/I_{\rm sat0}$ for dip-iv: five-level model prediction (dashed black line) and experimental data (coloured dots) for vapour cell temperatures 24.2 °C, 36.5 °C and 53.5 °C at a beam diameter $\sim$2.86 mm. The solid black line denotes the two-level saturation $\frac{1}{\sqrt{1+I/I_{\rm sat0}}}$.

(or equivalently decrease in absorption) even for probe intensities much smaller than $I_{\rm sat0}$, indicating a reduction in effective saturation intensity as compared to that expected from the two-level model. This is due to the transfer of some of the atomic population from the two-level system (coupled with probe) to other 'dark' hyperfine ground state. Due to the large hyperfine splitting between the two ground levels ($\sim$3 GHz for ^{85}Rb and $\sim$6.8 GHz for ^{87}Rb, much greater than the Doppler linewidth of $\sim$500 MHz at room temperature), atoms transferred into the dark state are now far-detuned from resonance and do not couple to the probe laser. This lowering of the atomic population in the coupled ground state manifests as reduced probe absorption [37].

4.2. Probe absorption as a function of beam diameter

Our five-level rate equation model also correctly predicts the dependence of probe absorption on the beam diameter and vapour cell temperature, both of which affect the transit relaxation rate (see equation (9)). Here, we focus on dip iii and iv that are significantly affected by hyperfine-pumping. In order to visualize the lowering of saturation intensity as compared to the simple two-level model, we calculated the normalised line-centre absorption coefficient, $\frac{\alpha_{\rm D0}(I)}{\alpha_{\rm D0}(I\to 0)}$, as a function of $I/I_{\rm sat0}$. Here, $\alpha_{\rm D0}$ denotes the maximum absorption coefficient of a particular dip at line-centre.

The five-level model correctly predicts (matches the experimental data) the lowering of saturation intensity (as compared to $I_{\rm sat0}$) with increasing beam diameter (see figure 5). According to equation (9), the transit time (or $\sim 1/\Gamma_{\rm t}$) scales linearly with the probe beam diameter, and increasing beam size leads to enhanced hyperfine pumping due to the greater atom–laser interaction time. A significant lowering of the effective saturation intensity is observed, as large as $\sim 10\times$ as compared to $I_{\rm sat0}$ for beam diameters of $\sim 6-7$ mm (see tables 2 and 3).

Now, if the beam diameter exceeds the value beyond which transit time becomes greater than the optical pumping time, then further increase in the beam diameter should not affect the probe absorption too much (see table 3). In other words, further increase in the interaction time will have negligible effect as the atomic system has already evolved to close to its steady-state. It is also observed in table 3 that for higher beam diameter, the standard deviation in the estimated value of β increases. This is because of the approximation of the actual complicated transit-probability function [18] by a simple exponential ($e^{-\Gamma_{\rm t} t}$) [4]. However, the exponential still shows better agreement with the exact expression [18] as compared to other reported approximations [22, 34], and is easy to incorporate in the rate equation model without a significant increase in error (as validated with experimental data). Another approximation in our calculation is to take the probe intensity as a constant within the beam cross-section i.e. assuming a 'top-hat' intensity profile. This works well (as validated by comparison with experimental data) for probe intensities up to $\sim 10 I_{\rm sat0}$, as can be inferred from [38]. In addition, our five-level model is also validated with the experimental results for a wide range of vapour cell temperature, ranging from $\sim$24 °C to 53 °C (where optical depth is $\lesssim 1$). For even higher probe intensities or at higher vapor cell temperatures (where optical depth is $\gg 1$), the spatial dependence of absorption saturation comes into picture and can be incorporated by multiplying the constant intensity (I) with the appropriate transverse profile (Gaussian in our case) and then averaging over the position coordinates [38].

4.3. Probe absorption as a function of vapour temperature

In [3], a comparison between absolute theoretically calculated (via the two-level χ-model) and experimentally observed Doppler-broadened transmission spectra of the Rb D_2-line for three different vapour cell temperatures at sufficiently weak probe $\sim 0.002 I_{\rm sat0}$ is given. As expected, a decrease in transmission (or increase in absorption) is observed for increasing vapour cell temperature. This is because absorption is directly proportional to the vapour atomic density, which increases drastically with temperature [23, 24]. The vapour atomic density (n) of both isotopes of Rb at a given temperature can be calculated by using the expressions given in [3, 32]. In figure 6(a), we show similar dependence of the probe transmission spectrum on vapour cell temperature, but at a higher probe intensity

$\gtrsim 0.1 I_{sat0}$ (where hyperfine pumping starts taking effect) that is captured by the five-level model quite well.

The five-level model also correctly predicts that the variation in vapour cell temperature in the vicinity of room temperature has a negligible effect on the value of effective I_{sat} or β (the reduction factor in saturation intensity compared to I_{sat0}). It can be seen that the normalized line-centre absorption vs I/I_{sat0} calculated using the five-level rate equation model for dip-iv at three different vapour cell temperatures basically all fall on the same curve (see figure 6(b)), in good agreement with experimental observations. This can be explained as transit time $(1/\Gamma_t)$ is inversely proportional to the square root of vapour cell temperature (i.e., $\sqrt{T}$). Here, the unit of T is Kelvin (=273 + T in °C), and at the Kelvin scale, the variation of the square root of the vapour cell temperature for ∼10 °C − 30 °C change will be insignificant. Hence any variation in the vapour cell temperature, on this scale, does not visibly affect the absorption saturation.

5. Conclusion

In conclusion, we calculate the Doppler-broadened transmission spectra of the Rb D_2-line using a five-level rate equation model for a broad set of experimental parameters and verify its validity through extensive measurements. We demonstrate that the five-level model correctly predicts up to a ∼10× lower saturation intensity than the basic two-level model, induced by hyperfine pumping. Unlike Cs, Rb has both its isotopes stable [14]. Also, the D_2-line comprises both closed and open hyperfine transitions, whereas the D_1-line consists of only open transitions. In addition, hyperfine levels in the excited state of Rb D_2-line are closely separated (∼a few 10 s–100 s of MHz) as compared to Cs D_1-line (∼a few GHz) and lie within one Doppler width. As indicated in [39], if the excited hyperfine levels interval is smaller than the full homogeneous linewidth (comprising of the natural linewidth, transit-time linewidth, pressure broadening (e.g. due to the presence of a buffer gas), and power broadening) of allowed transitions to these states from a given hyperfine ground state, then they should not be treated independently. For example, in ^{85}Rb ($F_g = 2 \rightarrow F_e = 1, 2, 3$), corresponding to dip-iii indicated above, the frequency interval between $F_e = 1$ and 2 is 29.260 MHz (∼5× natural linewidth) and becomes comparable to the full homogeneous linewidth at high probe intensities ($I > I_{sat0}$) due to power-broadening (note that the intensity threshold would be higher for the other dips i, ii, and iv, as the frequency intervals between the hyperfine excited states are larger). At such high probe intensities, optical pumping occurs simultaneously via both these excited levels for a particular velocity class of atoms at a given laser detuning. Thus, the five-level model, treating these excited states simultaneously, seems more adequate than the three-level model (see figure 7). Also, if the probe diameter is so narrow that the transit-time linewidth exceeds the hyperfine energy level separation, then the five-level model would be required even for lower probe intensities ($I < I_{sat0}$). All these factors combined make the Rb D_2-line less amenable to a three-level model as compared to the Cs D_1-line [3, 4]. Thus, we extend the basic three-level model for hyperfine pumping to a more general five-level model by including all the dipole allowed transitions ($\Delta F = 0, \pm 1$). We demonstrate that our model is able to accurately reproduce (with rms error <5%) absorption spectra over a wide range of experimental parameters, like probe beam diameter (∼1 − 6 mm), vapour cell temperature (∼24 °C − 53 °C) and probe intensity (up to ∼10I_{sat0}), for a linearly polarized probe and in the absence of any applied magnetic field. Thus, the full multi-level rate equation model (including all magnetic sub-levels) with the incorporation of actual transit-probability function (much more computationally intensive) [4, 18, 31] is not really required for the aforementioned experimental conditions.

In conclusion, our *ab initio* five-level model can predict accurate Doppler lineshapes for any given experimental parameter set for a linearly polarized probe without any fitting parameters, and can be easily applied to any other atomic system by an appropriate change of the atomic constants in the rate equations. However, for circular and elliptical polarized probes, Zeeman optical pumping within the magnetic sub-levels (along with the hyperfine optical pumping) also comes into the picture, which is beyond the scope of our five-level model (and is taken into account by the full multi-level rate equation model and/or the density matrix method at the cost of a ∼100× increase in computational time/resources) [25, 26, 39–41]. Although, even with a linearly (π) polarized probe, for hyperfine transition $F \rightarrow F - 1$, the atomic population gets transferred into the 'dark' $m_f = \pm F$ sub-levels in $|a\rangle$ ground states via Zeeman optical pumping; whereas for transition $F \rightarrow F$, $m_f = 0$ acts as a 'dark' sub-level in $|a\rangle$ ($\Delta m_f = 0$ is not allowed for $\Delta F = 0$ according to the electric-dipole selection rule) [2]. Thus in both these cases, there are some 'dark' sub-levels in $|a\rangle$ apart from the dark hyperfine level $|c\rangle$ (note that for transition $F \rightarrow F + 1$, there are no such additional 'dark' sub-levels as all the ground Zeeman levels in $|a\rangle$ are optically coupled to the excited states). For systems like the Cs D_1 line, where the excited hyperfine levels ($F - 1$ and F) are separated by more than the Doppler width, this Zeeman pumping (and resulting 'dark' sub-levels) induces measurable absorption saturation even for a linearly polarized probe [4]. On the other hand, for the Rb D_2 line, the three allowed excited states ($F - 1, F$ and $F + 1$) lie within a single Doppler width. Here, for dips i and ii (see figure 3), the closed hyperfine transitions $F \rightarrow F + 1$, (having relative coupling strength much greater than the other two open hyperfine transitions $F \rightarrow F - 1, F$) overwhelm the effect of Zeeman pumping as well as hyperfine pumping to some extent, whereas for dips iii and iv, hyperfine pumping (due to open transitions $F \rightarrow F, F + 1$) plays a key role and dominates over Zeeman pumping (due to closed transition $F \rightarrow F - 1$) for linearly polarized light [40].

Finally, the five-level model shows that even a weak probe, having intensity much less than I_{sat0}, can bring some changes in the effective ground state population in a multi-level system due to optical pumping. This information would be helpful in pump–probe spectroscopy (polarization and saturated absorption spectroscopy) to differentiate the effect of the pump from

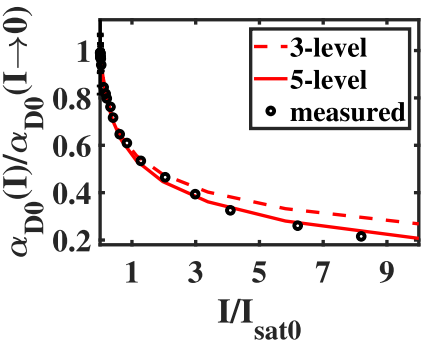


Figure 7. Comparison of three-level and five-level models with the experimentally observed normalized line-centre absorption coefficient for dip-iii of the Doppler-broadened transmission spectra of the Rb D_2 line at beam diameter ∼1.39 mm and vapour cell temperature 27.2 °C. It is evident that both models provide similar results for probe intensities $\lesssim I_{sat0}$. Beyond this, the result calculated from the three-level model starts deviating from the five-level model. Clearly, the five-level model shows better agreement with the experimental data, supporting the argument given in [39].

the probe on the medium [3, 37]. Also, the study of hyperfine-induced absorption saturation can be helpful in modelling nonlinear phenomena like optical switching, polarization-ellipse rotation, four-wave mixing, light-drag experiments, coupling of the atomic state dynamics with the axial and radial evolution of probe beam intensity, etc [12, 38, 42, 43]. The effect of hyperfine pumping can also be used to create an optical delay in thermal atomic vapour [44–46] which may have applications in the field of optical and quantum communications [42, 47].

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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